

TABLE 81 **LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)**

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.91	1.01	1.11	1.22	1.35	1.49	1.65	1.83	10.5%
Rest of Latin America	1.03	1.14	1.25	1.37	1.51	1.66	1.84	2.04	10.3%
Total	1.95	2.15	2.36	2.60	2.86	3.15	3.48	3.87	10.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

9.4 OTHER END USERS

This segment is inclusive of the application of microbiome sequencing services in clinical research, diagnostics, and collaborative R&D activities conducted by small-scale CROs, CDMOs, hospitals, and clinics. Biomarkers provided by the human gut microbiome may allow a new means to distinguish between a patient's physiology and disease conditions. It can also enable mapping microbial fingerprints and defining microbial-derived biomarkers that can help in identifying individuals at high risk of diseases, ensuring the diagnosis of existing conditions, and monitoring the efficacy of treatment.

For example, a bacterial microbiota diversity analysis—used to check for differences in the strain and number of microbes in chronic kidney disease patients and healthy people—found that there was an abundance of Bacteroidetes and Proteobacteria and fewer Firmicutes as compared to a healthy individual. Consequently, this provided a marker for the diagnosis of CKD. Furthermore, there are some diseases that occur because of the translocation of microbes from their normal habitat into some other tissues, where they can trigger diseases by setting up an inflammatory cascade. For example, it was found in a study that lupus, an autoimmune disease, was caused because of the translocation of *Enterococcus gallinarum* from the gut to the liver and other tissues, which triggered an autoimmune response. These are two instances of the potential use of the human microbiome as a diagnostic tool.

Several firms are focusing on investigation in this area. Enterome SA, for example, raised USD 13.8 million in venture capital funding for developing tests that use the composition of gut bacteria to diagnose inflammatory and liver diseases. It also entered into an agreement with Bristol-Myers Squibb for the discovery and development of microbiome-derived biomarkers, drug targets, and bioactive molecules to be developed as potential companion diagnostics and therapeutics for cancer. This will greatly support the prospects of the human microbiome in clinical research.

TABLE 82 MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER END USERS, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	28.83	32.57	36.79	41.58	47.05	53.41	60.85	69.62	13.6%
Europe	15.73	17.84	20.22	22.94	26.05	29.69	33.94	38.98	14.0%
Asia Pacific	4.37	5.06	5.85	6.77	7.84	9.10	10.59	12.36	16.1%
Latin America	1.00	1.09	1.19	1.31	1.43	1.56	1.72	1.89	9.6%
Middle East and Africa	0.69	0.75	0.81	0.87	0.93	1.01	1.08	1.17	7.7%
Total	50.62	57.31	64.87	73.48	83.30	94.76	108.18	124.03	13.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 83 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER END USERS, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	26.12	29.54	33.39	37.78	42.78	48.61	55.42	63.47	13.7%
Canada	2.70	3.03	3.39	3.80	4.27	4.80	5.43	6.15	12.6%
Total	28.83	32.57	36.79	41.58	47.05	53.41	60.85	69.62	13.6%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 84 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER END USERS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	3.39	3.87	4.41	5.03	5.74	6.57	7.55	8.72	14.6%
UK	3.04	3.48	3.98	4.55	5.22	6.00	6.92	8.01	15.0%
France	2.40	2.73	3.10	3.52	4.00	4.56	5.22	6.00	14.2%
Italy	0.68	0.77	0.87	0.98	1.11	1.26	1.44	1.65	13.7%
Spain	0.52	0.59	0.66	0.75	0.85	0.96	1.09	1.24	13.3%
Rest of Europe	5.70	6.41	7.21	8.12	9.15	10.34	11.73	13.36	13.1%
Total	15.73	17.84	20.22	22.94	26.05	29.69	33.94	38.98	14.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis